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I'm currently pursuing Data Science, and I'm interested in AI and technology. I'm a curious person who likes learning new things and building projects.

Description

A role-based Single-Page Web Application (SPA) that automates campus recruitment activities for Students, Companies, and the Placement Admin.

Technologies used

1. Backend & Flask Extensions (Python)

Flask (v2.3.3)

Flask-SQLAlchemy (v3.1.1)

Flask-JWT-Extended (v4.5.3)

Flask-CORS (v4.0.0)

Flask-Caching (v2.1.0)

Werkzeug Security

2. Frontend & UI Libraries (Javascript / Vue)

Vue 3 (v3.3.4)

Vite (v4.4.9)

Vue Router (v4.2.4)

Axios (v1.5.0)

Bootstrap 5 & Bootstrap Icons:

Responsive CSS layout engine providing pre-styled utility classes, grids, buttons, badges, popup modals, and iconography.

Chart.js & vue-chartjs (v4.3.3): Data visualization library used to render interactive Doughnut and Bar analytics charts on the Admin Dashboard.

3. Database & Storage

SQLite3:

Lightweight, zero-configuration relational database embedded directly into the application directory (placement_portal.sqlite3).

4. Asynchronous Processing & Background Tasks

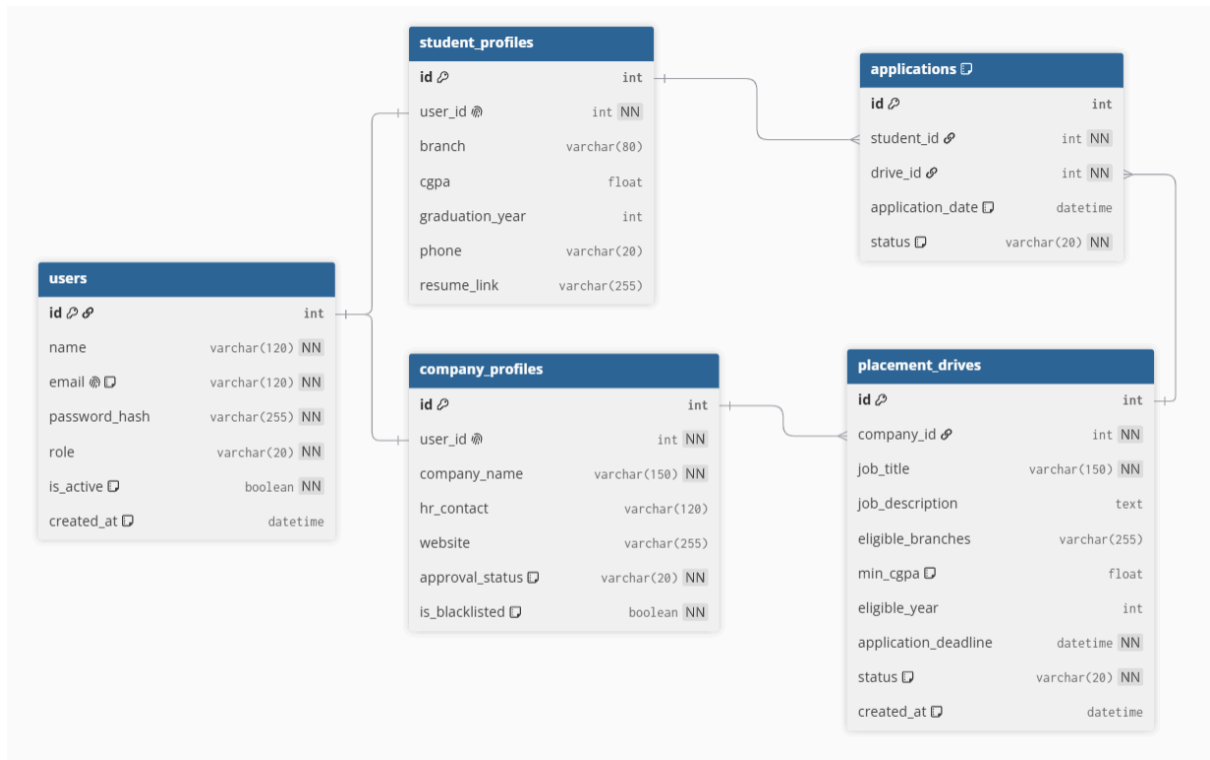
Celery (v5.3.4)

Distributed task queue used for user-triggered async jobs (e.g. CSV history export) and scheduled periodic tasks (e.g. daily deadline reminders & monthly reports).

Redis (v5.0.0)

In-memory data storage engine used as the Celery Message Broker and Result Backend.

DB Schema Design



API Design

The application implements a RESTful JSON API built with Flask Blueprints (/api/*) to handle data exchange between the Vue 3 frontend and the SQLite database.

Architecture and Features

The project follows a decoupled Single-Page Application (SPA) architecture. The controllers (API Endpoints) reside in backend/routes/*.py as Flask Blueprints, while database models live in backend/models.py. The views and templates reside in frontend/src/views/*.vue as Vue 3 components, styled with Bootstrap 5. Flask (app.py) handles API routing and serves the compiled Vue frontend (frontend/dist/).

Additional Features:

Offer Letter Generator: Company generates printable offer letter previews for selected candidates.

Interactive Analytics: Chart.js Doughnut and Bar charts for drive and application statistics.

Static ATS Resume Preview: Simulated keyword matching score preview for students.